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MATHEMATICS - 2007

Objective Questions: (Q.1-Q.15)

- (1.) Let $A(t)$ denote the area bounded by the curve $y = e^{-|x|}$, the x -axis and the straight lines $x = -t$ and $x = t$. Then $\lim_{t \rightarrow \infty} A(t)$ is equal to
- (a.) 2
 - (b.) 1
 - (c.) $\frac{1}{2}$
 - (d.) 0
- (2.) If k is a constant such that $xy + k = e^{(x-1)^2/2}$ satisfies the differential equation
- $$x \frac{dy}{dx} = (x^2 - x - 1)y + (x - 1),$$
- then k is equal to
- (a.) 1
 - (b.) 0
 - (c.) -1
 - (d.) -2

(3.) Which of the following functions is uniformly continuous on the domain as stated?

(a.) $f(x) = x^2, x \in \mathbf{R}$

(b.) $f(x) = \frac{1}{x}, x \in [1, \infty)$

(c.) $f(x) = \tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(d.) $f(x) = [x], x \in [0, 1]$

($[x]$ is the greatest integer less than or equal to x)

(4.) Let R be the ring of polynomials over $\mathbf{Z}_2$ and let I be the ideal of R generated by the polynomial $x^3 + x + 1$. Then the number of elements in the quotient ring $\frac{R}{I}$ is

(a.) 2

(b.) 4

(c.) 8

(d.) 16

(5.) Which of the following sets is a basis for the subspace $W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t, y + t = 0 \right\}$ of the vector space of all real 2×2 matrices?

(a.) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

(b.) $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

(c.) $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$

(d.) $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

(6.) Let G be an Abelian group of order 10. Let $S = \{g \in G : g^{-1} = g\}$. Then the number of non-identity elements in S is

(a.) 5

(b.) 2

(c.) 1

(d.) 0

(7.) Let (a_n) be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is

divergent. Let $S_n = \sum_{k=1}^n a_k$ for $n = 1, 2, \dots$ and

$t_n = \sum_{k=2}^n \frac{a_k}{S_{k-1}S_k}$ for $n = 2, 3, \dots$. Then $\lim_{n \rightarrow \infty} t_n$ is equal to

(a.) $\frac{1}{a_1}$

(b.) 0

(c.) $\frac{1}{(a_1 + a_2)}$

(d.) $a_1 + a_2$

(8.) For every function $f : [0, 1] \rightarrow \mathbf{R}$ which is twice differentiable and satisfies $f'(x) \geq 1$ for all $x \in [0, 1]$, we must have

(a.) $f''(x) \geq 0$ for all $x \in [0, 1]$

(b.) $f(x) \geq x$ for all $x \in [0, 1]$

(c.) $f(x_2) - x_2 \leq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$
with $x_2 \geq x_1$

(d.) $f(x_2) - x_2 \geq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$
with $x_2 \geq x_1$

(9.) Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Which of the following statements holds regarding the continuity and the existence of partial derivatives of f at $(0, 0)$?

(a.) Both partial derivatives of f exists at $(0, 0)$
and f is continuous at $(0, 0)$

- (b.) Both partial derivatives of f exists at $(0,0)$ and f is NOT continuous at $(0,0)$
- (c.) One partial derivatives of f does NOT exist at $(0,0)$ and f is continuous at $(0,0)$
- (d.) One partial derivatives of f does NOT exists at $(0,0)$ and f is NOT continuous at $(0,0)$

(10.) Suppose (c_n) is a sequence of real numbers such that $\lim_{n \rightarrow \infty} |c_n|^{1/n}$ exists and is non-zero. If the radius of convergence of the power series $\sum_{n=0}^{\infty} c_n x^n$ is equal to r , then the radius of

convergence of the power series $\sum_{n=1}^{\infty} n^2 c_n x^n$ is

- (a.) Less than r
- (b.) Greater than r
- (c.) Equal to r
- (d.) Equal to 0

(11.) The rank of the matrix $\begin{bmatrix} 1 & 4 & 8 \\ 2 & 10 & 22 \\ 0 & 4 & 12 \end{bmatrix}$ is

- (a.) 3
- (b.) 2
- (c.) 1

(d.) 0

(12.) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a continuous function.

If $\int_0^x f(2t) dt = \frac{x}{\pi} \sin(\pi x)$ for all $x \in \mathbf{R}$, then

$f(2)$ is equal to

(a.) -1

(b.) 0

(c.) 1

(d.) 2

(13.) Let $\vec{u} = (ae^x \sin y - 4x)\hat{i} + (2y + e^x \cos y)\hat{j} + az\hat{k}$,

where a is constant. If the line integral $\oint_C \vec{u} \cdot d\vec{r}$

over every closed curve C is zero, then a is equal to

(a.) -2

(b.) -1

(c.) 0

(d.) 1

(14.) One of the integrating factors of the differential

equation $(y^2 - 3xy)dx + (x^2 - xy)dy = 0$ is

(a.) $\frac{1}{(x^2 y^2)}$

(b.) $\frac{1}{(x^2 y)}$

(c.) $\frac{1}{(xy^2)}$

(d.) $\frac{1}{(xy)}$

(15.) Let C denote the boundary of the semi-circular disk $D = \{(x, y) \in \mathbf{R}^2 : x^2 + y^2 \leq 1, y \geq 0\}$ and let $\phi(x, y) = x^2 + y$ for $(x, y) \in D$. If $\hat{n}$ is the outward unit normal to C , then the integral $\oint_C (\vec{\nabla} \phi) \cdot \hat{n} \, ds$, evaluated counter-clockwise over C is equal to

(a.) 0

(b.) $\pi - 2$

(c.) π

(d.) $\pi + 2$

Subjective Questions: (Q.16-Q.29)

(16.) (a.) Let $M = \begin{bmatrix} 1+i & 2i & i+3 \\ 0 & 1-i & 3i \\ 0 & 0 & i \end{bmatrix}$.

Determine the eigenvalues of the matrix $B = M^2 - 2M + I$.

(b.) Let N be a square matrix of order 2. If the determinant of N is equal to 9 and the sum of the diagonal entries of N is equal to 10, then determine the eigenvalues of N .

(17.) (a.) Using the method of variation of parameters, solve the differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2, \text{ given that } x \text{ and } \frac{1}{x} \text{ are two solutions of the corresponding homogenous equation.}$$

(b.) Find the real number α such that the differential equation

$$\frac{d^2 y}{dx^2} + 2(\alpha - 1)(\alpha - 3) \frac{dy}{dx} + (\alpha - 2)y = 0$$

has a solution $y(x) = a \cos(\beta x) + b \sin(\beta x)$ for some non-zero real numbers a, b, β .

(18.) (a.) Let a, b, c be non-zero real numbers such that $(a - b)^2 = 4ac$.

Solve the differential equation

$$a(x + \sqrt{2})^2 \frac{d^2 y}{dx^2} + b(x + \sqrt{2}) \frac{dy}{dx} + cy = 0$$

(b.) Solve the differential equation

$$dx + (e^{y \sin y} - x)(y \cos y + \sin y) dy = 0$$

(19.) Let $f(x, y) = x(x - 2y^2)$ for $(x, y) \in \mathbf{R}^2$. Show that f has a local minimum at $(0, 0)$ on every straight line through $(0, 0)$. Is $(0, 0)$ a critical point of f ? Find the discriminant of f at $(0, 0)$. Does f have a local minimum at $(0, 0)$? Justify your answers.

(20.) (a.) Find the finite volume enclosed by the paraboloids $z = 2 - x^2 - y^2$ and $z = x^2 + y^2$.
 (b.) Let $f : [0, 3] \rightarrow \mathbf{R}$ be a continuous function

with $\int_0^3 f(x) dx = 3$.

Evaluate $\int_0^3 [x f(x) + \int_0^x f(t) dt] dx$

(21.) (a.) Let S be the surface

$$\{(x, y, z) \in \mathbf{R}^3 : x^2 + y^2 + 2z = 2, z > 0\}$$

and let $\hat{n}$ be the outward unit normal to S . If $\vec{F} = y\hat{i} + xz\hat{j} + (x^2 + y^2)\hat{k}$, then

evaluate the integral $\int_S \vec{F} \cdot \hat{n} dS$.

(b.) Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$. If a scalar field ϕ and a vector field $\vec{u}$ satisfy $\vec{\nabla}\phi = \vec{\nabla} \times \vec{u} + f(r)\vec{r}$ where f is an arbitrary

differentiable function, then show that $\nabla^2 \phi = rf'(r) + 3f(r)$.

- (22.) (a.) Let D be the region bounded by the concentric spheres $S_1 : x^2 + y^2 + z^2 = a^2$ and $S_2 : x^2 + y^2 + z^2 = b^2$, where $a < b$. Let $\hat{n}$ be the unit normal to S_1 directed away from the origin. If $\nabla^2 \phi = 0$ in D and $\phi = 0$ on S_2 , then show that

$$\iiint_D |\vec{\nabla} \phi|^2 dV + \iint_{S_1} \phi (\vec{\nabla} \phi) \cdot \hat{n} dS = 0$$

- (b.) Let C be the curve in $\mathbb{R}^3$ given by $x^2 + y^2 = a^2$, $z = 0$ traced counter-clockwise, and let $\vec{F} = x^2 y^3 \hat{i} + \hat{j} + z \hat{k}$. Using Stokes' theorem, evaluate $\oint_C \vec{F} \cdot d\vec{r}$.

- (23.) Let V be the subspace of $\mathbf{R}^4$ spanned by the vectors $(1, 0, 1, 2)$, $(2, 1, 3, 4)$ and $(3, 1, 4, 6)$. Let $T : V \rightarrow \mathbf{R}^2$ be a linear transformation given by $T(x, y, z, t) = (x - y, z - t)$ for all $(x, y, z, t) \in V$. Find a basis for the null space of T and also a basis for the range space of T .

- (24.) (a) Compute the double integral $\iint_D (x+2y) dx dy$, where D is the region in the xy -plane bounded by the straight lines $y = x + 3$, $y = x - 3$, $y = -2x + 4$ and $y = -2x - 2$.

(b.) Evaluate: $\int_0^{\pi/2} \left[\int_{\pi/2}^{\pi} \frac{\sin x}{x} dx \right] dy + \int_{\pi/2}^{\pi} \left[\int_y^{\pi} \frac{\sin x}{x} dx \right] dy$

- (25.) (a.) Does the series $\sum_{k=1}^{\infty} \frac{(-1)^k k + x^k}{k^2}$ converges uniformly for $x \in [-1, 1]$? Justify.

- (b.) Suppose (f_n) is a sequence of real-valued functions defined on $\mathbf{R}$ and f is a real-valued function defined on $\mathbf{R}$ such that $|f_n(x) - f(x)| \leq |a_n|$ for all $n \in \mathbf{N}$ and $a_n \rightarrow 0$ as $n \rightarrow \infty$. Must the sequence (f_n) be uniformly convergent on $\mathbf{R}$? Justify.

- (26.) (a.) Suppose f is a real-valued thrice differentiable function defined on $\mathbf{R}$ such that $f'''(x) > 0$ for all $x \in \mathbf{R}$. Using Taylor's formula, show that

$$f(x_2) - f(x_1) > (x_2 - x_1) f' \left(\frac{x_1 + x_2}{2} \right)$$

for all x_1 and x_2 in $\mathbf{R}$ with $x_2 > x_1$.

(b.) Let (a_n) and (b_n) be sequences of real numbers such that $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ for all $n \in \mathbf{N}$. Must there exist a real number x such that $a_n \leq x \leq b_n$ for all $n \in \mathbf{N}$? Justify your answer.

(27.) Let G be the group of all 2×2 matrices with real entries with respect to matrix multiplication. Let G_1 be the smallest subgroup of G containing $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, and G_2 be the smallest subgroup of G containing $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Determine all elements of G_1 and find their orders. Determine all elements of G_2 and find their orders. Does there exist a one-to-one homomorphism from G_1 onto G_2 ? Justify.

(28.) (a.) Let p be a prime number and let $\mathbf{Z}$ be the ring of integers. If an ideal J of $\mathbf{Z}$ contains the set $p\mathbf{Z}$ properly, then show that $J = \mathbf{Z}$.

[Here $p\mathbf{Z} = \{px : x \in \mathbf{Z}\}$].

(b.) Consider the ring $R = \{a + ib : a, b \in \mathbb{Z}\}$ with usual addition and multiplication. Find all invertible elements of R .

- (29.) (a.) Suppose E is a non-empty subset of $\mathbf{R}$ which is bounded above, and let $\alpha = \sup E$. If E is closed, then show that $\alpha \in E$. If E is open, then show that $\alpha \notin E$.
- (b.) Find all limit points of the set

$$E = \left\{ n + \frac{1}{2m} : n, m \in \mathbf{N} \right\}$$



Rajendra Dubey (Known as Dubey Sir): He has experience of almost three decades of teaching mathematics. During his journey, he has guided IIT-JEE, IAS, NET, GATE, IIT-JAM aspirants etc. Apart from producing top rankers in the various exams, he is widely known for making mathematics interesting and fascinating. He is founder & director of DIPS ACADEMY and chairman of Dubey's Group of Companies.



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